



Lithium Tris[3-fluoro-1,2-benzenediolato(2-)-O,O']phosphate as a Novel Lithium Salt for Lithium Battery Electrolytes

Noritoshi Nanbu, Koji Tsuchiya, Takatsugu Shibazaki, and Yukio Sasaki^{*,z}

Department of Applied Chemistry, Faculty of Engineering, Tokyo Institute of Polytechnics, Atsugi, Kanagawa 243-0297, Japan

We have synthesized a novel lithium salt with a chelate compound of phosphorus, lithium tris[3-fluoro-1,2-benzenediolato(2-)-O,O']phosphate (3-FLTBP). The thermal stability and electrolytic properties of 3-FLTBP, and the coulombic efficiency of the deposition/dissolution cycling for a lithium metal (lithium electrode cycling efficiency) are investigated. 3-FLTBP is more thermally stable than lithium tris[1,2-benzenediolato(2-)-O,O']phosphate (LTBP) and lithium tris[4-methyl-1,2-benzenediolato(2-)-O,O']phosphate (4-MLTBP), which we previously synthesized. The electrochemical stability in propylene carbonate solutions and the specific conductivities in ethylene carbonate-dimethyl carbonate (mole ratio 1:1) binary solutions decrease in the following ascending order: 3-FLTBP, LTBP, and 4-MLTBP. The mixing of the lithium organophosphate and LiPF₆ enhances the cycling efficiency over the range of high cycle numbers.

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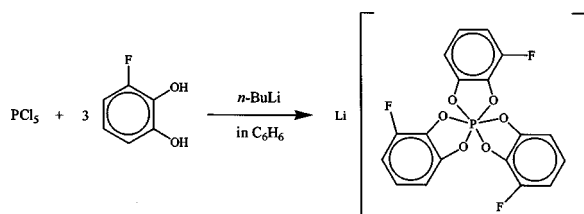
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Growing interest has been focused on the development of novel lithium salts containing fluorine atoms and on their application to electric vehicles, because solutions containing the lithium salts show high conductivity and oxidation durability. Lithium salts with chelate compounds are known to be highly thermally stable as compared to common lithium salts. Barthel *et al.* reported several lithium salts with chelate compounds of boron¹⁻⁶ such as lithium bis[3-fluoro-1,2-benzenediolato(2-)-O,O']borate. Recently, Xu and Angell have also synthesized lithium organoborates of a weakly coordinating anion^{7,8} such as lithium bis[1,2-tetra(trifluoromethyl)ethylenediolato(2-)-O,O']borate. The use of various chelating ligands enables us to control the electron density around a central atom in complexes and to investigate the properties systematically. We previously synthesized chelate compounds of phosphorus, lithium tris[1,2-benzenediolato(2-)-O,O']phosphate (LTBP)⁹ and lithium tris[4-methyl-1,2-benzenediolato(2-)-O,O']phosphate (4-MLTBP).¹⁰ The thermal stability, the electrolytic properties, and the discharge or charge/discharge characteristics were investigated. The mixing of 4-MLTBP and LiPF₆ markedly increases the cycling efficiency over the range of high cycle numbers when compared to the efficiency observed in a single electrolyte system of LiPF₆.

In this paper we report the thermal characteristic and electrolytic behavior of lithium tris[3-fluoro-1,2-benzenediolato(2-)-O,O']phosphate (3-FLTBP), which we have newly synthesized. The coulombic efficiency of the deposition/dissolution cycling for a lithium metal (lithium electrode cycling efficiency) is also described.

Experimental

Synthesis.—3-FLTBP has been synthesized according to the procedure for the synthesis of LTBP,⁹ except that a pentacoordinate compound, which is an intermediate in the reaction, was not isolated from the solution



Scheme I.

* Electrochemical Society Active Member.

^z E-mail: sasaki@chem.t-kougei.ac.jp

Phosphorus pentachloride 13.8 g (0.06 mol) was dissolved in benzene (350 mL) boiling in a beaker under an inert atmosphere, and 3-fluorocatechol 25.0 g (0.18 mol) was carefully added to the solution with stirring. A hexane solution of *n*-butyllithium 10 mL (0.016 mol) was added dropwise to the solution and then a product was precipitated. The product was filtered off, purified by recrystallization from 1,2-dimethoxyethane (DME), and dried *in vacuo* at 140°C for 48 h. The total yield of the product was 40%. The results of elemental analyses including inductively coupled plasma (ICP) emission spectroscopy were in good agreement with the calculated values; found: C, 51.1; H, 2.85; F, 12.6; Li, 1.65; P, 7.42%, and calculated: C, 51.9; H, 2.16; F, 13.5; Li, 1.67; P, 7.44%. 3-FLTBP was also identified by means of ¹H-¹³C correlation spectroscopy (COSY) and ³¹P nuclear magnetic resonance (NMR). These NMR spectra in acetone-d₆ containing 1% tetramethylsilane (TMS) were recorded on a JEOL JNM-LA500 Fourier transform-NMR system. Figure 1 shows a ¹H-¹³C COSY NMR spectrum of 3-FLTBP in acetone-d₆ containing 1% TMS. Peaks marked with symbols can be assigned to protons and carbons in 3-FLTBP. For the measurement of ³¹P NMR spectra, 85 wt% D₃PO₄ was used as the external reference. A signal of ³¹P was shifted downfield to -77.24 ppm in the spectrum of 3-FLTBP when compared to the signal for LTBP (-81.71 ppm).

Apparatus and measurements.—Thermogravimetry was carried out on a Seiko Instruments, Inc., thermogravimetric differential thermal analysis (TG/DTA)220 system with a sample mass of *ca.* 10 mg. The apparatus and techniques for measurements of specific conductivity,^{9,11,12} viscosity,¹³ and voltammograms^{9,12} were reported previously. The coulombic efficiency of the deposition/dissolution cycling for a lithium metal (lithium electrode cycling efficiency) was measured with a 6-channel charge/discharge unit (Hokuto Denko, model HJ-101SM6) and estimated by a galvanostatic plating/stripping method reported by Koch and Brummer.¹⁴ Ethylene carbonate (EC) and dimethyl carbonate (DMC) used were battery grade and dehydrated by purified molecular sieves (4A) before preparation of solutions.

Results and Discussion

Thermal stability.—Thermal stability of 3-FLTBP was investigated by thermogravimetry and compared with that of LTBP and 4-MLTBP. Figure 2 shows TG curves of 3-FLTBP, 4-MLTBP, and LTBP measured in air. As the temperature increased, the masses of 3-FLTBP, 4-MLTBP, and LTBP started to decrease at *ca.* 250, 160, and 150°C, respectively. This finding indicates that the thermal decomposition temperature decreases in that order. The order agreed with that of the molecular weights. The molecular weights of

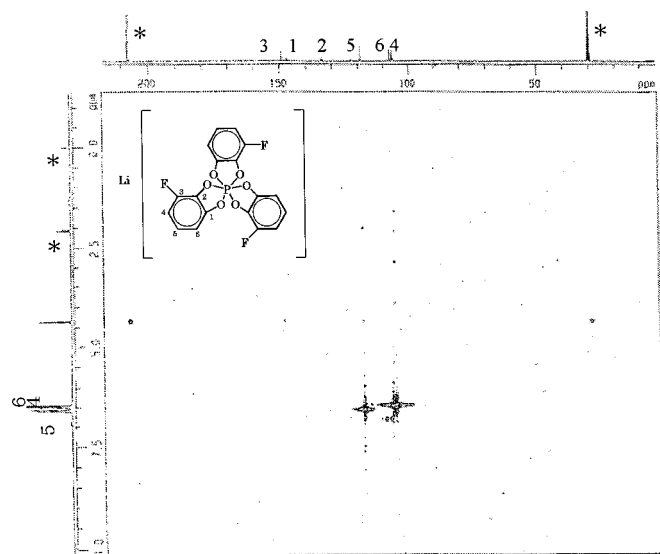


Figure 1. ^1H - ^{13}C COSY NMR spectrum of 3-FLTBP in acetone- d_6 containing 1% tetramethylsilane. Peaks marked with symbols can be assigned to protons and carbons in 3-FLTBP. Peaks marked with * are ascribable to acetone- d_6 .

3-FLTBP, 4-MLTBP, and LTBP are 416.2, 404.3, and 366.2, respectively. The organophosphate anions contain bidentate ligands and are highly thermally robust because of the chelate effect. Furthermore, the substituent effect was observed for 3-FLTBP and 4-MLTBP, which have heavier anions, similarly to lithium salts with chelate compounds of boron.¹⁵ The inductive and resonance effects of a fluorine atom can thermally stabilize the chelate anion of 3-FLTBP, because the fluorine atom is highly electronegative and has lone pairs of electrons. In the case of 4-MLTBP, the hyperconjugation arising from the overlap of σ -orbitals in the methyl group with the π -system in the benzene ring can increase the thermal stability of the chelate anion.¹⁰

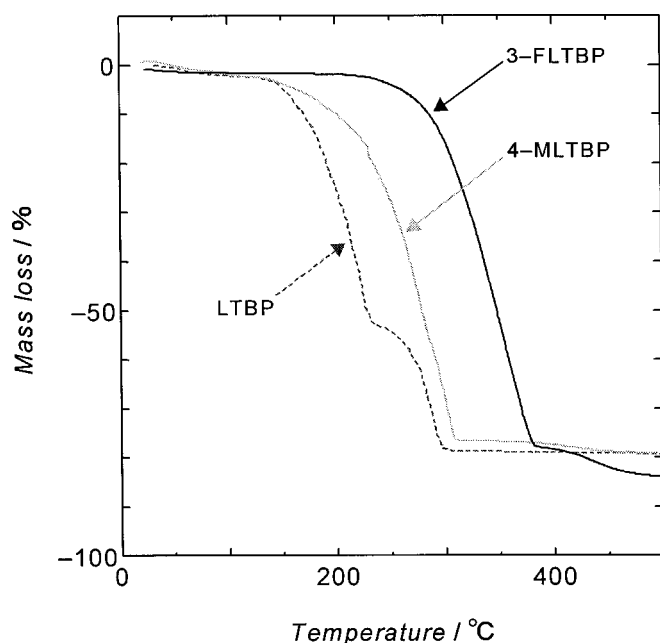


Figure 2. Thermogravimetric curves of 3-FLTBP, 4-MLTBP, and LTBP at a heating rate of 1 K min^{-1} in air. α -Alumina was used as the reference.

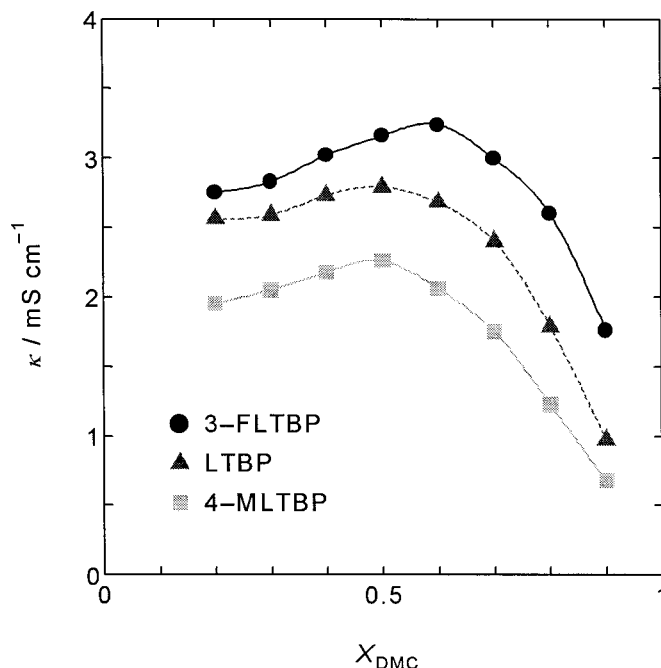


Figure 3. Specific conductivities (κ) of 0.5 mol dm^{-3} 3-FLTBP, 4-MLTBP, and LTBP in EC-DMC binary solutions at 25°C as functions of a mole fraction of DMC (X_{DMC}). The concentration of the electrolyte was 0.5 mol dm^{-3} .

Electrolytic properties.—Figure 3 shows the variations in specific conductivity (κ) of 0.5 mol dm^{-3} LTBP, 4-MLTBP, and 3-FLTBP in EC-dimethyl carbonate (DMC mole ratio 1:1) binary solutions. X_{DMC} refers to a mole fraction of DMC. Since the solubilities of the chelate compounds in the binary solvent were less than 1 mol dm^{-3} , the concentration of the lithium salt was adjusted to 0.5 mol dm^{-3} . It is well known that the blending of a high polarity solvent and a low viscosity solvent results in the synergistic effect on conductivity. The electrolyte solution shows that the conductivity is higher than that of either single solvent system. The conductivities of the chelate compounds decreased in approximately the following ascending order: 3-FLTBP, LTBP, and 4-MLTBP. This fact can be explained in terms of the difference in the degree of electrolytic dissociation of the lithium organophosphates. A fluorine atom, which has powerful electron-withdrawing ability, disperses the charge on the anion. This idea is supported by the finding that a NMR signal of ^{31}P in 3-FLTBP (-77.24 ppm) appears in a down-field position as compared to that for LTBP (-81.71 ppm). Furthermore, in the case of 3-FLTBP, the value of X_{DMC} at which the conductivity showed a maximum was higher than that for the other lithium organophosphates. Therefore, the degree of electrolytic dissociation of 3-FLTBP can be higher than that for LTBP. Specific conductivity is affected by ionic mobility of a lithium salt, which relates to viscosity in the electrolyte solution, as well as the degree of electrolytic dissociation of the lithium salt. Table I shows viscosities (η) of 3-FLTBP, 4-MLTBP, LTBP, and equimolar mixtures of the lithium organophosphate and LiPF_6 in EC-DMC (mole ratio 1:1) binary solutions. The concentration of the single or the mixed lithium salt was adjusted to 0.5 mol dm^{-3} . The order of the viscosities was essentially the same as that of anion sizes: 4-MLTBP, 3-FLTBP, and LTBP. However, since the concentration of 3-FLTBP dissociating into ions in the solution is higher than those for the other lithium organophosphates, the introduction of fluorine atoms can lead to enhanced conductivity in spite of the increase in viscosity. This finding plays an important role in the design of electrolytes for lithium batteries. On the other hand, a low degree of electrolytic

Table I. Specific conductivities (κ) and viscosities (η) in EC-DMC (mole ratio 1:1) binary solutions containing LTBP, 4-MLTBP, 3-FLTBP, or equimolar mixtures of the chelate compound and LiPF_6 at 25°C. The concentration of the single or the mixed lithium salt was adjusted to 0.5 mol dm^{-3} .

Electrolytes	κ (mS cm^{-1})	η (mPa s)
LTBP	2.62	2.826
4-MLTBP	2.25	3.142
3-FLTBP	3.16	2.939
LTBP + LiPF_6	5.66	1.876
4-MLTBP + LiPF_6	5.26	2.068
3-FLTBP + LiPF_6	5.73	2.025

dissociation of 4-MLTBP, which is ascribable to the electron-releasing effect of methyl groups, decreases the conductivity in co-operation with the high viscosity.¹⁰

Table I also shows specific conductivities of LTBP, 4-MLTBP, 3-FLTBP, and equimolar mixtures of the chelate compound and LiPF_6 in EC-DMC (mole ratio 1:1) binary solutions. The conductivities of 3-FLTBP, 4-MLTBP, and LTBP were much lower than that of $0.5 \text{ mol dm}^{-3} \text{ LiPF}_6$, 9.66 mS cm^{-1} . However, such lower conductivities are improved by mixing with LiPF_6 .

The delocalization of electrons in the chelate anion of 3-FLTBP also accounts for the higher oxidative decomposition potential, *ca.* 4.1 V vs. Li/Li^+ , as compared to LTBP (*ca.* 3.7 V vs. Li/Li^+) and the corresponding lithium organoborate (lithium bis[3-fluoro-1,2-benzenediolato(2-)-*O,O'*]borate, 3.75 V vs. Li/Li^+).^{3,4} Furthermore, this value is comparable to that for the perfluorinated lithium organoborate (lithium bis[tetrafluoro-1,2-benzenediolato(2-)-*O,O'*]borate, 4.1 V vs. Li/Li^+).^{3,4} Figure 4 shows linear potential sweep voltammograms of 0.1 mol dm^{-3} lithium organophosphates in propylene carbonate solutions. A potential reached at 0.5 mA cm^{-2} was regarded as the oxidative decomposition potential of the lithium salt. We consider that the decomposition potentials of the lithium organophosphates may not be appreciably dependent on the concentration at 0.1 mol dm^{-3} or higher. On the other hand, the

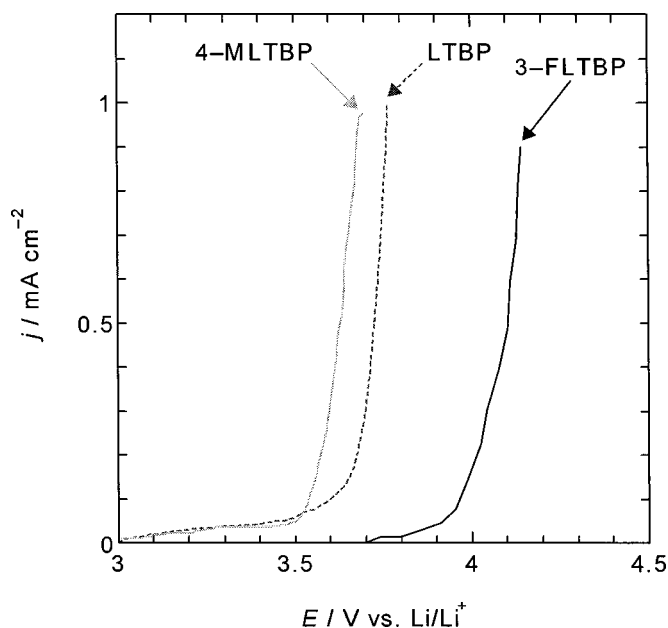


Figure 4. Linear potential sweep voltammograms of 0.1 mol dm^{-3} LTBP, 4-MLTBP, and 3-FLTBP in PC solutions on a Pt electrode at 25°C. Scan rate: 10 mV s^{-1} .

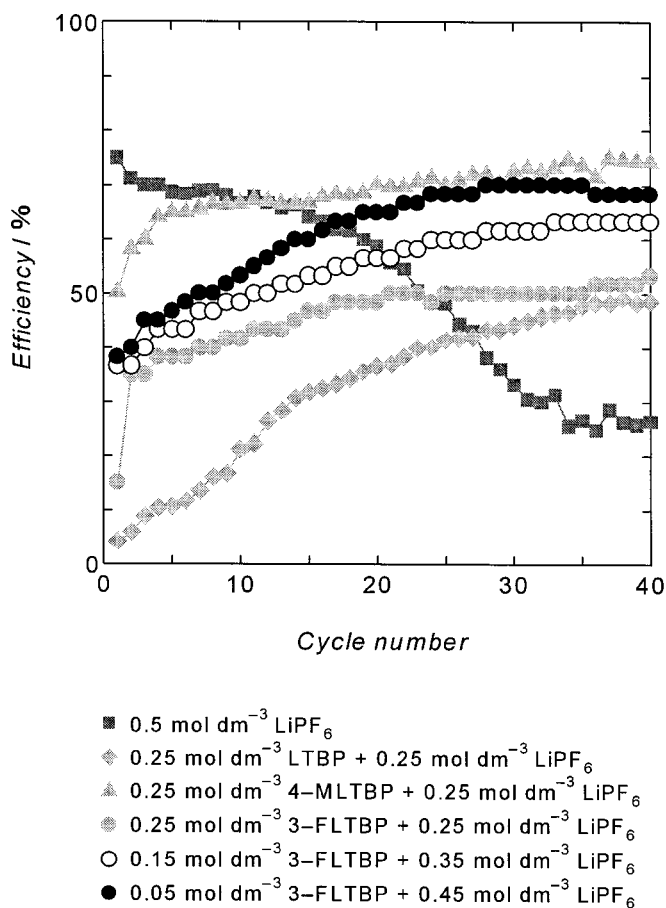


Figure 5. Variations in the coulombic efficiency of the deposition/dissolution cycling for a lithium metal in EC-DMC (mole ratio 1:1) binary solutions containing LiPF_6 , LTBP, 4-MLTBP, 3-FLTBP, or mixtures of the lithium organophosphate and LiPF_6 at 25°C. The concentration of the single or the mixed electrolyte was adjusted to 0.5 mol dm^{-3} . The plating and the stripping current density were 1 mA cm^{-2} , and the plated charge density was 300 mC cm^{-2} .

localization of electrons in 4-MLTBP makes its oxidative decomposition potential lower than those of LTBP.¹⁰ The voltage limits of a mixture can be governed by a component that shows a lower decomposition potential. The oxidative decomposition potentials of the lithium organophosphates were considerably lower than that of LiPF_6 , which was more than 5.5 V vs. Li/Li^+ . Accordingly, one would expect that the oxidative decomposition potential of a mixture of the lithium organophosphate and LiPF_6 should be determined by that of the lithium organophosphate.

Cycling efficiency.—Figure 5 shows the variations in coulombic efficiency of the deposition/dissolution cycling for a lithium metal (lithium electrode cycling efficiency) in EC-DMC (mole ratio 1:1) binary solutions containing $0.5 \text{ mol dm}^{-3} \text{ LiPF}_6$ or several binary electrolytes. It was difficult to measure the cycling efficiency in solutions containing the single lithium organophosphate, because the efficiency decreased drastically with increasing the cycle number. Thus, mixtures of the lithium organophosphate and LiPF_6 were used as the electrolytes. The cycling efficiency in the solution containing the equimolar mixture of 3-FLTBP and LiPF_6 was lower than that for the equimolar mixture of 4-MLTBP and LiPF_6 . However, the cycling efficiency increased with increasing a mole ratio of LiPF_6 to 3-FLTBP. In the case of a 9:1 mole ratio, the cycling efficiency was highly stable and *ca.* 70% over the range of high cycle numbers. The

electrolyte solution of LiPF_6 showed the lowest cycling efficiency over that range. The mixing of the LTBP derivative and LiPF_6 can improve the morphology, thickness, and chemical composition of a surface film formed on a Ni working electrode, so that high reversibility in the deposition and dissolution of a lithium metal is observed even over the range of high cycle numbers.

For solutions containing a single lithium organoborate, the cycling efficiency is very high even over the range of high cycle numbers,^{12,15,16} though the conductivities do not differ from those in solutions of the lithium organophosphate too greatly. Hence, the cycling efficiency is essentially independent of conductivity in the electrolyte solution.^{12,16} The morphology, thickness, and chemical composition of the surface film formed in the solution containing the lithium organophosphate due to reduction of reactive impurities and electrolytes may be different from those for the lithium organoborate.

Conclusions

We have newly synthesized 3-FLTBP. The substituent effect on the thermal stability and the electrolytic properties was observed. 3-FLTBP is more thermally and electrochemically stable than LTBP and 4-MLTBP. The solutions containing 3-FLTBP showed higher conductivity in spite of its high viscosity, because the degree of electrolytic dissociation of 3-FLTBP is higher. The mixing of the LTBP derivative and LiPF_6 enhanced the cycling efficiency over the range of high cycle numbers. Surface analyses can clarify the

relationship between the cycling efficiency and the morphology, thickness, and chemical composition of the surface film.

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